

System Project: Technical Requirements (2026 SM2)

Key details

Read the respective section below for full details.

Section Title	Key Details
Team Composition	A team of 6-9 members with some disciplinary diversity.
Sub-team composition	Sub-teams of size 2 are the base case. Teams with an odd number of members may have one sub-team of size 1 or 3.
Timing	Starts in Week 8 of semester and concludes in the exam period.
Subject Value	In total the System Project contributes 50% to the overall subject grade
Project Definition	The team themselves define the details of their particular project, to fit within the guideline described on this page. A video of the lane-change behaviors from 2024 is shown in this section (video from 2025 is still in production).
Project Goal	Make a robotic system (or other engineering system) autonomously and reliably perform exciting behaviors.
Sub-team Goals and Operation	As part of achieving the overall project goal, each sub-team focuses on different aspects and the whole team works together to integrate all those aspects into achieving the overall project goal.
Guidance for setting and managing sub-team goals	Sub-teams are a defining aspect of a System Project! Hence, setting and managing sub-team goals is part of the challenge and the learning experience of a System Project. This section provides guidance based on observations from 2025, for example, create a block diagram for the whole system your team needs to build, and assign one block to each sub-team.
How to avoid a “flop”	This so-called "System Project" was first introduced to the subject in 2025 and in 2026 we are making it even better (hopefully!), based on experiences & feedback from 2025.

Raison d'être:

The ELEN90095 subject's "raison d'être" is for everyone to have a great System Project learning experience in an interdisciplinary team. So please talk to the teaching team about any aspects of teamwork and project work (be they positive, negative, or neutral aspects).

Team Composition

You self-select your teams and must meet the following requirements:

- Team size 6-9.
- At least 2 different master's programs across the team members (and preferably spanning some IT/CS and some Eng team members).

Forming a team across the two workshop sessions is allowed:

- If you think that you would like to do this, please contact the teaching team.
- Workshops for Weeks 10-12 will be essentially one session from 10am-4pm.
- This is why forming a team across workshop sessions is allowed.
- As a baseline plan, your whole team should attend together.
- Hence, the teaching team will only approve such teams where all team members are available, as a minimum, for the same 3 hours out of the 10am-4pm window on Tuesdays.

Sub-team Composition

As per the table at top of page, sub-team composition requirements are:

- Sub-teams of size 2 are the base case.
- Teams with an odd number of members may have one sub-team of size 1 or 3.
- Speak with the teaching team if your preferred sub-teams don't meet these requirements.

See the "Sub-team Goals and Operation" section below for a description of what sub-teams actually do.

Timing

Week 8 Lecture slot

- Talk through the details of the System Project and of the hardware platform we are using for the System Project.

	Attendance is NOT mandatory, and watching the lecture recording before the end of the week is mandatory.
Week 8 Workshop session	- Activities to facilitate team formation and to facilitate generating a (first draft) project goal for your team. Attendance is mandatory for this Week 8 workshop session.
Week 9 Lecture slot	The Technical Content Test (https://canvas.lms.unimelb.edu.au/courses/238681/pages/technical-content-test-details-and-examples-2026-sm2) take places during this slot.
Week 9 Workshop session	- Activities to finalize team formation and to describe all the capabilities of the real-world robots we use for the project. Attendance is strongly encouraged for this Week 9 workshop session

Weeks 10-12 Workshops	- These will be essentially one session running 10am-4pm. - You do not have to attend the whole 6 hours. - As a baseline plan, your whole team should attend together. - In these sessions the teaching team goes around from team-to-team to discuss technical details and teamwork details. (And probably spends half-the-time debugging the hardware or software 🤖).
Attendance for Week 10-12 workshops	- Attendance is not mandatory for the Week 9, 10, and 12 workshop sessions. Attendance is mandatory for the Week 11 workshop session, as this is when we will conduct the System Project: Progress Assessment (https://canvas.lms.unimelb.edu.au/courses/238681/assignments/648926). - These are intended to be the "fun" weeks of working with real-world hardware :-) - We will be tracking attendance and checking-in with teams and team members that are absent.
Weeks 10-12 Lecture slots	Weeks 10-12: No lecture content. The class room is free for working on your System Project and teaching team will be there for consultation.

Exam Period	The System Project: Final Assessment (https://canvas.lms.unimelb.edu.au/courses/238681/assignments/648924) is
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conducted during the exam period.

Subject value

The 50% overall grade for the System Project is split across 3 assessments. The following table summarizes and the assignment pages are the "ground truth" for what the assessment is.

Grade	Title	Due Week	Br
15%	<u>System Project: Teamwork Submissions</u> (https://canvas.lms.unimelb.edu.au/courses/238681/assignments/648927)	Weekly submissions from Week 9 into the Exam Period	De as SM (ht pr
0%	<u>System Project: Progress Assessment</u> (https://canvas.lms.unimelb.edu.au/courses/238681/assignments/648926)	Week 11	Ar the
35%	<u>System Project: Final Assessment</u> (https://canvas.lms.unimelb.edu.au/courses/238681/assignments/648924)	Week 1-3 of Exam Period	A s tea • • Th

Project Definition

The team themselves define the details of their particular project!

The real-world robotic system that we have available in this subject is a 1:10 scale car that is equipped with sensors and computer so that you can make it perform autonomous behaviors.

A team **CAN** choose to focus on some other robotic system (or engineering system), for examples, renewables integration in the electric power grid.

The requirements for the project definition are:

- You define project goals that describe one or more system behavior that your team aims to achieve.

- **Achieving your project goals predominantly requires you to work on policies, algorithms, and code that are part of an AI4R block diagram.**
 - This **CAN** include code for the inner workings of a simulation.
 - This does **NOT** include physical build of a robot or parts of a robot.

In 2024, the project required every team to perform a lane-change with the 1:10 scale autonomous car and the project duration was 4 weeks in the middle of the semester. Here is a video collage of the system behaviors achieved.

In 2025, the System Project was very similar to this year. We haven't made a video collage yet :(Instead, here are two examples from the 2025 System Project Teams:

Project Goals (through examples)

The following is a list of **example aspects** to consider for a project that works with out 1:10 scale autonomous cars.

Note importantly that all items on this list are not equal:

- Some are algorithms;
- Some are behaviours
- Some are an overall project goals;
- Some are sub-team goals.

The intention of the list is to give your team **food-for-thought when you are defining the scope and details of the specific project goal for your team.**

List of example aspects for defining an overall project goal:

- **"Normal" driving behaviours:**
 - Smooth driving along the "road".
 - Smooth lane change.
 - Lane change around a stopped vehicle or obstacles.
 - Slalom between cones.
- **Road setup:**
 - Cones defining the road with blue on the left and yellow on the right side of the road.
 - Road boundaries with cones of all the same color.
 - Road boundaries with large variation in inter-cone spacing.
 - Road boundary with cones only on one side of the road.
 - Flexible pipes defining the road boundaries (which are detected by the Lidar).
- **Estimation:**
 - Using the time-correlation of subsequent cone detection measurements to improve car location estimates.
 - Detect whether another car is in our lane or another lane.
 - Estimating the "state" necessary as input to an MPC policy.
- **Policies:**
 - Cruise control lower-level policy.
 - Pure Pursuit policy.
 - MPC policy.
 - MPCC (Model Predictive Contouring Control) policy.
 - Deep RL policy.
 - Human-in-the-loop driving, i.e., the human is the primary driver of the car and your algorithms constantly check for imminent "crashes" and adjust the cars inputs to avoid the crash.
- **Parameter estimation:**
 - Experiments to identify important parameters of the real-world car (for example the mapping from steering request to turning radius).

- An automatic calibration procedure that identifies the steering offset (aka. the steering trim).
- **Anti-skidding behaviours:**
 - Anti-skid acceleration.
 - Anti-skid turning.
 - Emergency braking (anti-skid).
 - **Note: skidding behaviour, and hence anti-skidding policies, are more tricky than they might sound.**
- **Computer vision pipelines:**
 - **Note: Pursuing these aspects requires explicit approval of the teaching team, because without prior experience computer vision pipeline aspects are too time-consuming.**
 - Improved cone-detection pipeline.
 - A method to measure "ground truth" performance for the real-car.

Project aspects that are NOT allowed

These project aspects are **NOT** allowed to avoid you entering a rabbit hole of challenges that is not compatible with the robot quality or time available for this project:

- **SLAM (Simultaneous Localization And Mapping).** This is NOT allowed because the overall premise of the 1:10 cars is that they drove along an unseen track. Additionally, a full SLAM pipeline is computationally intensive and tends to consume a lot of time on implementation details that are not aligned with the goals of ELEN90095. That said, if you do aim to fuse cone-detections across time, then SLAM concepts and algorithms (e.g., FastSLAM2) can provide solid inspiration for this.
- **Physical build.** This is not allowed because it can be very time consuming and is not aligned with the focus on the ELEN90095 subject on policy algorithms.

Sub-team Goals and Operation

Operation:

During the project, the sub-teams:

- Work on aspects of the project that are distinct.
- Can change their composition.
- Can change the aspects they are working on.
- **Note importantly** that the aspects each sub-team works on are co-dependent by definition of being part of the same overall System Project goal (i.e., sub-teams work on sub-systems).

Goals:

A sub-team goal is an aspect of the project that 2 team members:

- Commit to working on together.
- Regularly inform the whole team about their progress and challenges.
- Work on until either:
 - That aspect is complete; or
 - The whole team together decides there is a need to re-define and re-allocate sub-team goals and workload.

If you are worried about “free riders”, we are too:

- In general, University projects with larger teams are:
 - **GREAT** because working in industry often involves working in larger teams.
 - **HORRIBLE** because there are many past experiences where a team member “free rides” and the others have to work harder to make up for it.
- By forcing you to explicitly nominate sub-teams of 2, we are intending to:
 - Create clearer accountability within your team; and
 - Achieve clearer visibility for us (i.e., for the teaching team) into how your teamwork is functioning.
- The assessment for the System Project is conducted as a team **and marked per individual**.
- We have designed the assessment to support collegial and collaborative teamwork, with the intention that:
 - **High-achieving individuals** can work on the project without feeling like they need the whole team to be as ambitious as they are.
 - **Low-contribution individuals** are flagged early to the teaching team, and their fellow team members help them find a way to contribute to the project (rather than trying to do extra work to cover up for them!).
 - **Every individual** has a fun, insightful, and positive experience working on the project!

Guidance for setting and managing sub-team goals:

Guidance #1: Sub-team goals are essentially a single block in the system-level block diagram:

- At the start of the project, get together as a team and **draw a detailed block diagram of what you need to create to achieve your overall project goal**.
- For example, this is likely to include identification of critical model parameters via experiments; estimate of the robot’s state; policy design and synthesis for performing the

desired behavior.

- **Allocate each sub-team one block** (or other critical aspect of the block diagram) as their goal to work on. In one simple sentence: **sub-teams work on sub-systems**.
- **Be sure to define the data types** (and frequency and requirements) being carried on every connection in your block diagram because this defines the agreed interface between algorithms and code of each sub-team.

Guidance #2: Two sub-teams can work on the same block or aspect with distinctly different approaches:

- Each choice for policy algorithms and architecture has **different pros and cons**.
- Core to the ELEN90095 subject is that you gain experience **comparing different policy algorithms and architectures** being applied to achieve the same type of system behavior (i.e., the Upskilling assignment is exactly this).
- Hence, for the System Project it is a valid, and **potentially very insightful**, to have two sub-teams working on the same block using distinctly different policy algorithms and/or architectures.
- For example, estimating the car state from cone detection data using: (A) various methods on each detection capture independently, **versus** (B) using a method that expects correlation between subsequent detection captures.
- Doing this can also be **a great way to reliably achieve the overall project goals** by choosing one of the approaches to be simpler and quicker to implement, thus freeing up that sub-team to work on a new sub-team goal, and thus getting a full working system sooner (albeit with mediocre performance).
- This suggestion is essentially a method to parallelize working on baseline and stretch goals for the project.

Guidance #3: Conducting tests of overall system behavior is NOT valid as the ONLY goal that a sub-team work on for the WHOLE project:

- It is true that in industry some sub-teams purely focus on conducting test of overall system behavior, and sub-system behaviors, without ever being involved in the development of the algorithms they are testing Or similarly a sub-team that works purely on system integration to achieve overall project specs. However, this is **NOT** compatible with the learning objective of ELEN90095.
- **Sub-teams should test their own developments:** Every individual and sub-team should work on designing, implementing, and testing at least one sub-system or aspect of the overall system.
- **Testing can occasionally be a pure sub-team goal:** As the project progresses, it can be valid to allocate a sub-team goal to be testing of a particular system behavior or aspect for a week or more.

Guidance #4: A “natural” flow of sub-team goals as the project progresses:

- **First half (e.g., first 3-4 weeks of the project):** allocate each sub-team a clearly defined block or aspect from the system-level block diagram. This gives every individual the required experience of designing, implementing, and testing an aspect of the overall system.
- **Then transition to an integration phase (remaining 2-3 weeks of the project):** What this integration phase looks like is highly dependent on what happens during the previous bullet point. For example:
 - (A) sub-teams could still be testing their “own stuff” but with a renewed focus on system-level integration; or
 - (B) sub-team composition is completely changed with the new sub-teams getting system-integration sub-goals than span multiple sub-systems; or
 - (C) anything in between.
- **The teamwork protocols should help:** the intention of the protocols and requirements that we “force” upon you (as per the [Teamwork Requirements page](https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-teamwork-requirements-2026-sm2) (<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-teamwork-requirements-2026-sm2>)) are intended to help with this “natural” flow of the System Project. Regular meetings and writing out of current sub-team goals and progress are necessary ingredients for your team to properly manage the “natural flow” of your System Project.

Guidance #5: The intended dynamic between sub-teams and the whole team:

- **Between members of a single sub-team** the work ethic should be fluid and flexible, readily adapting to technical challenges that arise while working towards their sub-team goals..
- **Across the whole team** the work ethic should be more formal, as guided by the [Teamwork Requirements page](https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-teamwork-requirements-2026-sm2) (<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-teamwork-requirements-2026-sm2>). Friction-at-first, and then eventual success, when defining interfaces between the sub-teams’ technical work and then trying to putting the pieces of work together into a whole system.

FAQ: Does having more team members mean the project is expected to achieve much better results or include more work?

- The team size does not correlated with expected project achievements in the way you might "naturally" expect.
- The aspect that is “constant” regardless of team size is the expected work of a sub-team.
- **As an example:**
 - A team of eight with 4 sub-teams (i.e., 2 people per sub-team) might choose to implement and compare 4 distinctly different approaches to following a track of cones

(with no obstacles or other environment complexities).

- In such a case, you might look at the project and say: “they didn’t achieve much for a team of 8; their car behavior can only follow cones and nothing else”.
- **We as the teaching team would look at this and say:**
 - We are not assessing based on the number of achieved behaviors, so implementing 4 distinctly different cone-following policy approaches is completely appropriate (See Guidance #2 above).
 - We are predominantly assessing the quality of each sub-team’s work towards their respective sub-team goal, so let’s get into the details of how each sub-team synthesized and tested each cone-following policy differently and how they coordinated sharing common learning and best practices between the sub-teams.

How to avoid a “flop”

Based on experience from 2025, the teaching team sees the most likely thing to make the System Project a flop as:

- Many individuals will be excited about many different topics.
- Hence, many individuals will want to work on all the exciting aspects of all the sub-teams.
- Hence, the idea of a team of 6 with three sub-teams working on three aspects in parallel, **devolves into 6 people all sitting around 1 laptop working through the project aspects sequentially.**
- Hence, the key experience of a system project of integrating sub-teams is completely lost.
- Hence, everyone leaves dissatisfied :-)

To avoid this scenario, we kindly ask the following of every individual:

- **Embrace the aspect your sub-team is working on to delve as deep into it as you can!**
- **Use team integration sessions to teach the other sub-teams about what you are working on.**
- **Regularly ask other sub-teams for their thoughts and ideas (especially when you feel stuck).**
- **This way everyone is still learning about multiple exciting aspects of the project!**

THANK YOU!